INCH-POUND
MIL-PRF-49467/2A
19 May 1997
SUPERSEDING
MIL-C-49467/2
5 April 1990

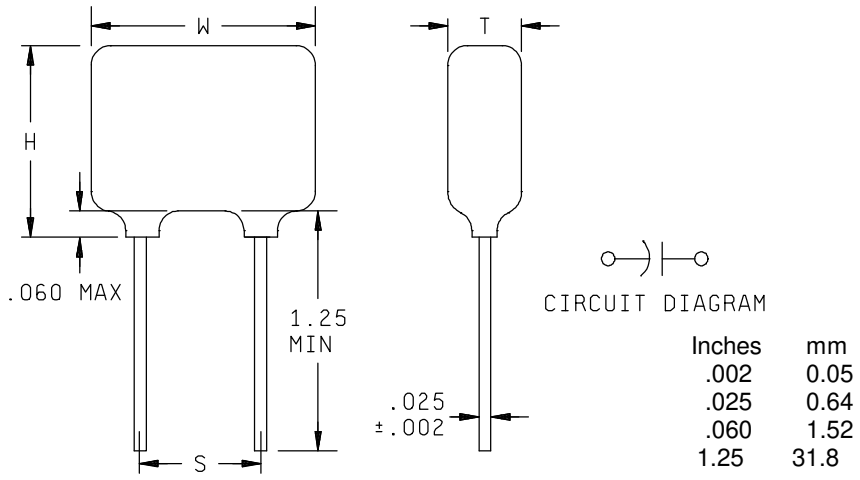
PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CERAMIC, MULTILAYER, HIGH VOLTAGE, 2,000 V DC, ESTABLISHED RELIABILITY, STYLE HVR02

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-49467.

*



Inches	mm
.002	0.05
.025	0.64
.060	1.52
1.25	31.8

Case code	Dimensions			
	W max	H max	T max	S ± .030 (0.76)
A	.250 (6.35)	.220 (5.59)	.200 (5.08)	.170 (4.32)
B	.320 (8.13)	.280 (7.11)	.250 (6.35)	.220 (5.59)
C	.370 (9.40)	.300 (7.62)	.250 (6.35)	.275 (6.98)
D	.470 (11.94)	.400 (10.16)	.270 (6.86)	.375 (9.52)
E	.570 (14.48)	.500 (12.70)	.270 (6.86)	.475 (12.06)
F	.670 (17.02)	.600 (15.24)	.270 (6.86)	.575 (14.60)
G	.770 (19.56)	.720 (18.29)	.270 (6.86)	.675 (17.14)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Capacitor style HVR02.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Conformally coated, molded, or encapsulated.

Lead material: In accordance with MIL-PRF-49467.

Capacitance value: See table I.

Capacitance tolerance: J = ±5 percent, K = ±10 percent, M = ±20 percent. See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: 2,000 V dc.

Failure rate level: M (1.0 percent), P (0.1 percent), R (.01 percent), and S (.001 percent).

Dielectric types: BP, BR, and BZ in accordance with MIL-PRF-49467.

Temperature coefficient: 0 ± 30 ppm/°C (BP), ±15 percent (BR, BZ).

* Voltage-temperature limits: In accordance with MIL-PRF-49467.

Symbol	Capacitance change with reference to +25°C	
	Steps A through D of voltage-temperature limit cycle table of MIL-PRF-49467. Bias = 0 volt.	Steps E through G of voltage-temperature limit cycle table of MIL-PRF-49467.
BP BR BZ	0 ± 30 ppm/°C ±15% ±15%	100% rated voltage: 0 ± 30 ppm/°C 100% rated voltage: +15, -40% 60% rated voltage: +15, -45%

Dissipation factor (+25°C): 2.5 percent maximum (BR and BZ characteristic) or 0.1 percent maximum (BP characteristic), measured under the same conditions as capacitance.

Resistance to soldering heat: In accordance with MIL-PRF-49467.

Partial discharge (corona): In accordance with MIL-PRF-49467.

* Radiographic inspection (molded and encapsulated case types only): In accordance with MIL-PRF-49467.

Part or Identifying Number (PIN). See table I.

Marking: In accordance with MIL-PRF-49467.

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* TABLE I. Capacitor characteristics.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
	pF			
M49467P02100--	10	J,K	BP	A
M49467P02120--	12	J,K	BP	A
M49467P02150--	15	J,K	BP	A
M49467P02180--	18	J,K	BP	A
M49467P02220--	22	J,K	BP	A
M49467P02270--	27	J,K	BP	A
M49467P02330--	33	J,K	BP	A
M49467P02390--	39	J,K	BP	A
M49467P02470--	47	J,K	BP	A
M49467P02560--	56	J,K	BP	A
M49467P02680--	68	J,K	BP	A
M49467P02820--	82	J,K	BP	A
M49467P02101--	100	J,K	BP	A
M49467R02101--	100	K,M	BR	A
M49467P02121--	120	J,K	BP	A
M49467R02121--	120	K,M	BR	A
M49467P02151--	150	J,K	BP	A
M49467R02151--	150	K,M	BR	A
M49467P02181--	180	J,K	BP	A
M49467R02181--	180	K,M	BR	A
M49467P02221--	220	J,K	BP	A
M49467R02221--	220	K,M	BR	A
M49467P02271--	270	J,K	BP	A
M49467R02271--	270	K,M	BR	A
M49467P02331--	330	J,K	BP	A
M49467R02331--	330	K,M	BR	A
M49467P02391--	390	J,K	BP	B
M49467R02391--	390	K,M	BR	A
M49467P02471--	470	J,K	BP	B
M49467R02471--	470	K,M	BR	A
M49467P02561--	560	J,K	BP	B
M49467R02561--	560	K,M	BR	A
M49467P02681--	680	J,K	BP	B
M49467R02681--	680	K,M	BR	A
M49467P02821--	820	J,K	BP	B
M49467R02821--	820	K,M	BR	A
M49467P02102--	1,000	J,K	BP	C
M49467R02102--	1,000	K,M	BR	A
M49467P02122--	1,200	J,K	BP	C
M49467R02122--	1,200	K,M	BR	A
M49467P02152--	1,500	J,K	BP	D
M49467R02152--	1,500	K,M	BR	A

See footnote at end of table.

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* TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
	pF			
M49467P02182--	1,800	J,K	BP	D
M49467R02182--	1,800	K,M	BR	B
M49467Z02182--	1,800	K,M	BZ	A
M49467P02222--	2,200	J,K	BP	D
M49467R02222--	2,200	K,M	BR	B
M49467Z02222--	2,200	K,M	BZ	A
M49467P02272--	2,700	J,K	BP	E
M49467R02272--	2,700	K,M	BR	B
M49467Z02272--	2,700	K,M	BZ	A
M49467P02332--	3,300	J,K	BP	E
M49467R02332--	3,300	K,M	BR	B
M49467P02392--	3,900	J,K	BP	E
M49467R02392--	3,900	K,M	BR	B
M49467P02472--	4,700	J,K	BP	E
M49467R02472--	4,700	K,M	BR	C
M49467Z02472--	4,700	K,M	BZ	B
M49467P02562--	5,600	J,K	BP	F
M49467R02562--	5,600	K,M	BR	C
M49467Z02562--	5,600	K,M	BZ	B
M49467P02682--	6,800	J,K	BP	F
M49467R02682--	6,800	K,M	BR	D
M49467Z02682--	6,800	K,M	BZ	B
M49467P02822--	8,200	J,K	BP	G
M49467R02822--	8,200	K,M	BR	D
M49467Z02822--	8,200	K,M	BZ	C
	µF			
M49467P02103--	.01	J,K	BP	G
M49467R02103--	.01	K,M	BR	D
M49467Z02103--	.01	K,M	BZ	C
M49467R02123--	.012	K,M	BR	D
M49467R02153--	.015	K,M	BR	D
M49467R02183--	.018	K,M	BR	E
M49467Z02183--	.018	K,M	BZ	D
M49467R02223--	.022	K,M	BR	E
M49467Z02223--	.022	K,M	BZ	D
M49467R02273--	.027	K,M	BR	E
M49467Z02273--	.027	K,M	BZ	D
M49467R02333--	.033	K,M	BR	E
M49467R02393--	.039	K,M	BR	F
M49467Z02393--	.039	K,M	BZ	E
M49467R02473--	.047	K,M	BR	F
M49467Z02473--	.047	K,M	BZ	E
M49467R02563--	.056	K,M	BR	G
M49467Z02563--	.056	K,M	BZ	E

See footnote at end of table.

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* TABLE I. Capacitor characteristics - Continued.

PIN 1/	Capacitance	Capacitance tolerance	Characteristic	Case code
	µF			
M49467R02683--	.068	K,M	BR	G
M49467Z02683--	.068	K,M	BZ	F
M49467R02823--	.082	K,M	BR	G
M49467Z02823--	.082	K,M	BZ	F
M49467R02104--	.10	K,M	BR	G
M49467Z02124--	.12	K,M	BZ	G
M49467Z02154--	.15	K,M	BZ	G

1/ The complete PIN shall include additional letter symbols to indicate the capacitance tolerance and failure rate level.

APPLICATION NOTES:

Additional encapsulation is necessary in applications where the possibility of a voltage breakdown between the leads of the capacitor, or between the capacitor and another potential, could occur.

Heat sinks on each lead or adequate preheating is required when these capacitors are installed in or removed from circuits by soldering iron.

Changes from previous issue: The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AT, AV, ME, MI
Navy - MC
Air Force - 19, 99
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

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